

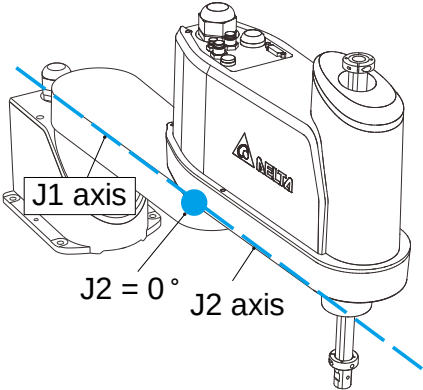
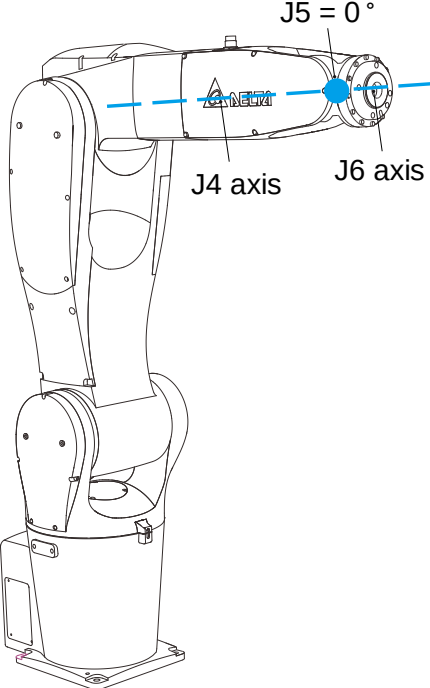
# Preface

This manual is written based on Delta Robotic Automation Studio (DRASudio) V1.00.07.55. Check for the version number before operating the software. If the version number is below V1.00.07.55, contact Delta and the distributor before operation.

## Safety precautions

### Singularities

Both SCARA and VA robots have singularities. The following describes singularity occurrence and avoidance.

| SCARA singularity                                                                                                                                                                                                                                                                                                                                                                                                                                                            | VA singularity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <p><b>Singularity occurrence:</b><br/>During robot movement, <math>J2 = 0^\circ</math> and the J1 axis and J2 axis become coincident. At this point, the J2 axis stops and cannot move.</p> <p><b>How to avoid singularities:</b></p> <ul style="list-style-type: none"><li>■ Move J2 by <math>\pm 5^\circ</math> to <math>15^\circ</math>.</li><li>■ Add multiple joint motions to avoid J2 from being <math>0^\circ</math>.</li><li>■ Use the MovP instructions.</li></ul> | <p><b>Wrist singularity occurrence:</b><br/>During robot movement, <math>J5 = 0^\circ</math> and the J4 axis and J6 axis become coincident. At this point, the J4 axis instantly rotates at a high speed; beware that the cables on the J4 axis are pulled along with the rotation.</p> <p><b>How to avoid singularities:</b></p> <ul style="list-style-type: none"><li>■ Move J5 by <math>\pm 5^\circ</math> to <math>15^\circ</math>.</li><li>■ Add multiple joint motions to avoid the J4 axis and J6 axis from becoming coincident.</li><li>■ Use the MovP instructions.</li></ul> |